

The power of adaptivity in representation learning: From meta-learning to federated learning

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Part 1: Overview and Few Shot Learning

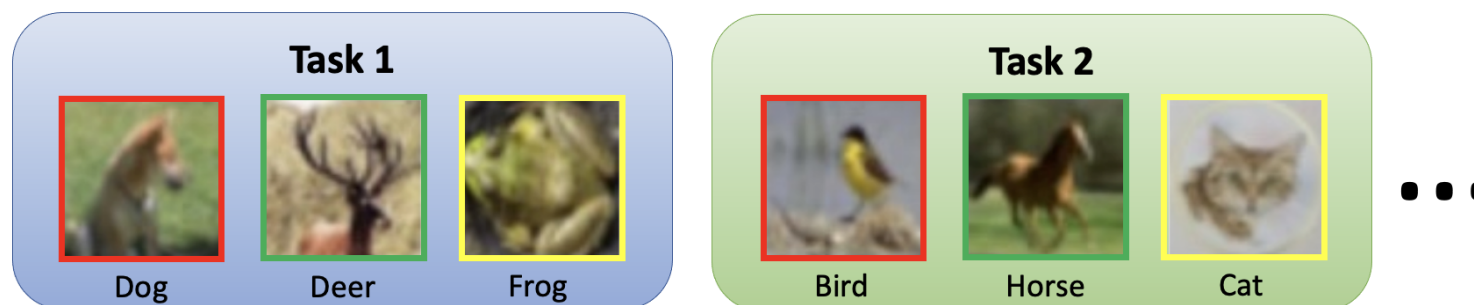
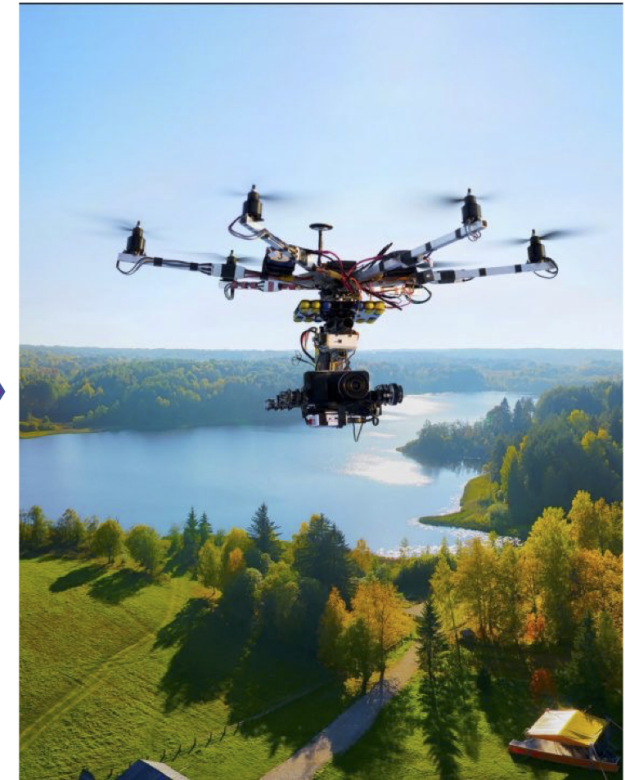


Image credits: Alex Krizhevsky, Learning Multiple Layers of Features from Tiny Images, 2009.

Central Goal: Generalize to New Tasks



Training Data



Task in new Scenario

Image Credits: bit.ly/3i5m8ay, bit.ly/3w723ZY, bit.ly/3KHMq5E, bit.ly/3i7pREJ, bit.ly/34I1ytT

- Training data in various scenarios; goal is to quickly adapt to the task in a new scenario

Few Shot Learning

- Suppose that we have: in aggregate, large amounts of data from many training environments and a small amount of data available just prior to deployment from the deployment environment
- Possible Approach:
 - ▶ (a) Build a model using data from the training environments
 - ▶ (b) Fine-tune the model using the small amount of deployment data

How can we build a model that is easily fine-tunable?

Obvious Approach: Build a model to minimize average training loss, and then fine tune for deployment

Average Risk Minimization (ARM) + Fine Tuning

- Set of tasks: $\mathcal{T} = \{\mathcal{T}_i\}_{i=1}^n$ coming from distribution p
- Select a model θ_{train}^*

Minimize Average Loss

$$\theta_{train}^* \in \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n f_i(\theta)$$

- A new task $\mathcal{T}_{test}$ is revealed, drawn according to dist. p
- Fine tune the model: $\theta_{train}^* \rightarrow \theta_{new}^*$
- Performance: $f_{test}(\theta_{new}^*)$

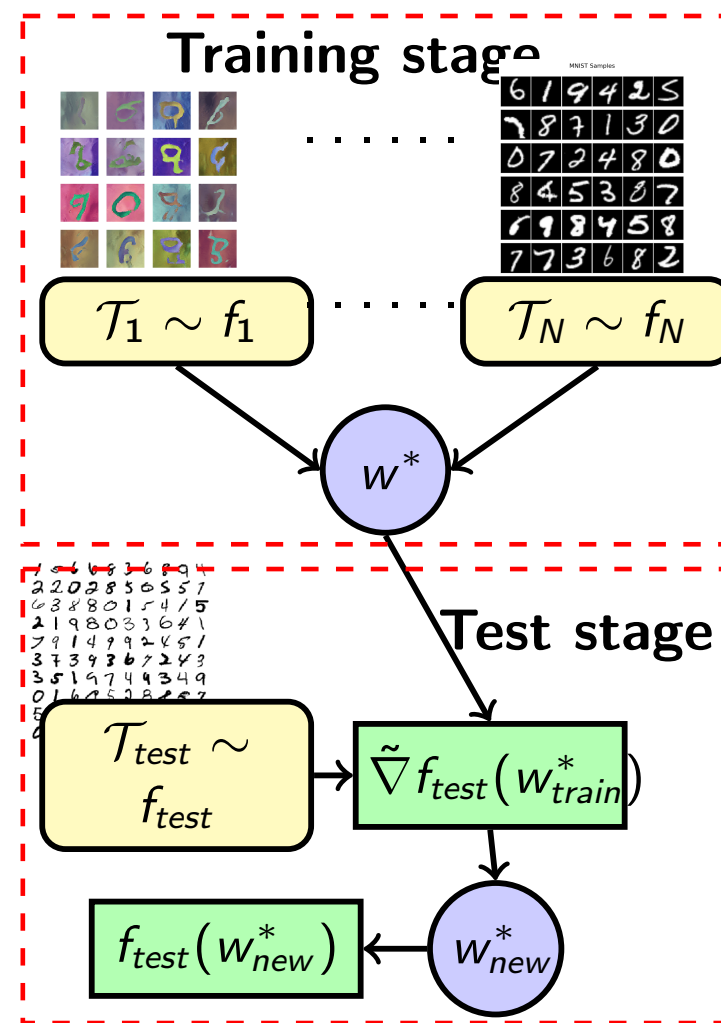
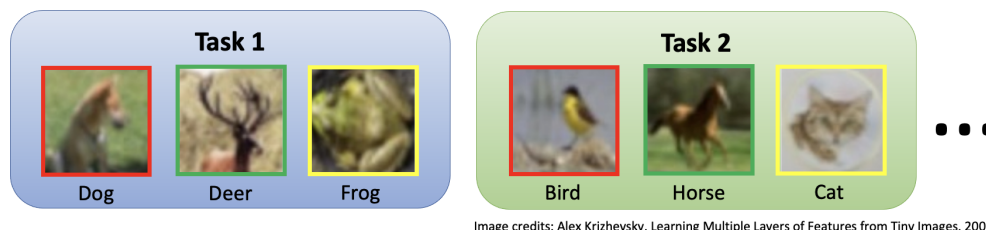


Image Credits: <https://bit.ly/392pda9>,

<https://bit.ly/3EEIElq>

Does (ARM + Fine Tuning) work well?

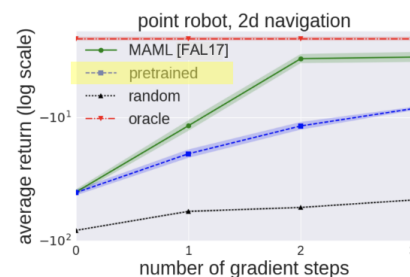
- Suppose we have images from a large number of classes (e.g., Imagenet)
 - ▶ Task == classifying images among a K -subset of these classes, small K
 - ▶ Many different subsets == Many tasks



- ARM + Fine Tuning has mixed performance [FAL17]

	5-way Accuracy	
	1-shot	5-shot
MiniImagenet (Ravi & Larochelle, 2017)		
fine-tuning baseline	$28.86 \pm 0.54\%$	$49.79 \pm 0.79\%$
nearest neighbor baseline	$41.08 \pm 0.70\%$	$51.04 \pm 0.65\%$
matching nets (Vinyals et al., 2016)	$43.56 \pm 0.84\%$	$55.31 \pm 0.73\%$
meta-learner LSTM (Ravi & Larochelle, 2017)	$43.44 \pm 0.77\%$	$60.60 \pm 0.71\%$
MAML, first order approx. (Finn et al., 2017)	$48.07 \pm 1.75\%$	$63.15 \pm 0.91\%$
MAML (Finn et al., 2017)	$48.70 \pm 1.84\%$	$63.11 \pm 0.92\%$

“Fine-tuning baseline”: Few-shot image classification
accuracy of ARM after fine-tuning (image taken from [FAL17])



“Pretrained”: Fine-tuning reward for ARM on robot
2d navigation task (image taken from [FAL17])

- ▶ (Aside: For multi-task linear regression, we can formally justify this)

Takeaway: ARM does not work well. Instead, researchers have ...

- Change the loss function (MAML)
- Change the training procedure (federated learning)

Part 2a: Changing the Loss Function – the MAML Approach

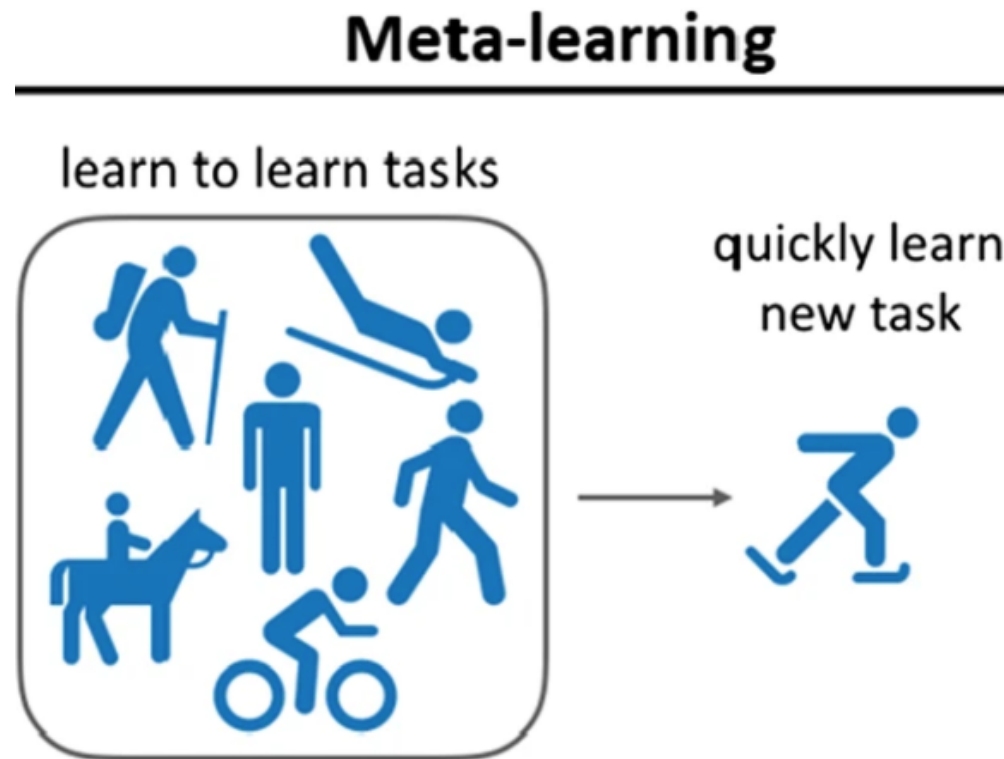


Image credit: <https://meta-world.github.io>, [HRJ21]

Model-Agnostic Meta Learning – Change the Loss Function

Model-Agnostic Meta Learning [FAL17]

- Set of tasks: $\mathcal{T} = \{\mathcal{T}_i\}_{i=1}^n$ coming from distribution p
- Select a model θ_{train}^*

Change the Loss Function

$$\theta_{train}^* \in \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n f_i(\theta - \alpha \nabla f_i(\theta; (X_i, y_i)))$$

- A new task $\mathcal{T}_{test}$ is revealed, drawn according to dist. p
- Fine tune the model: $\theta_{train}^* \rightarrow \theta_{new}^*$
- Performance: $f_{test}(\theta_{new}^*)$

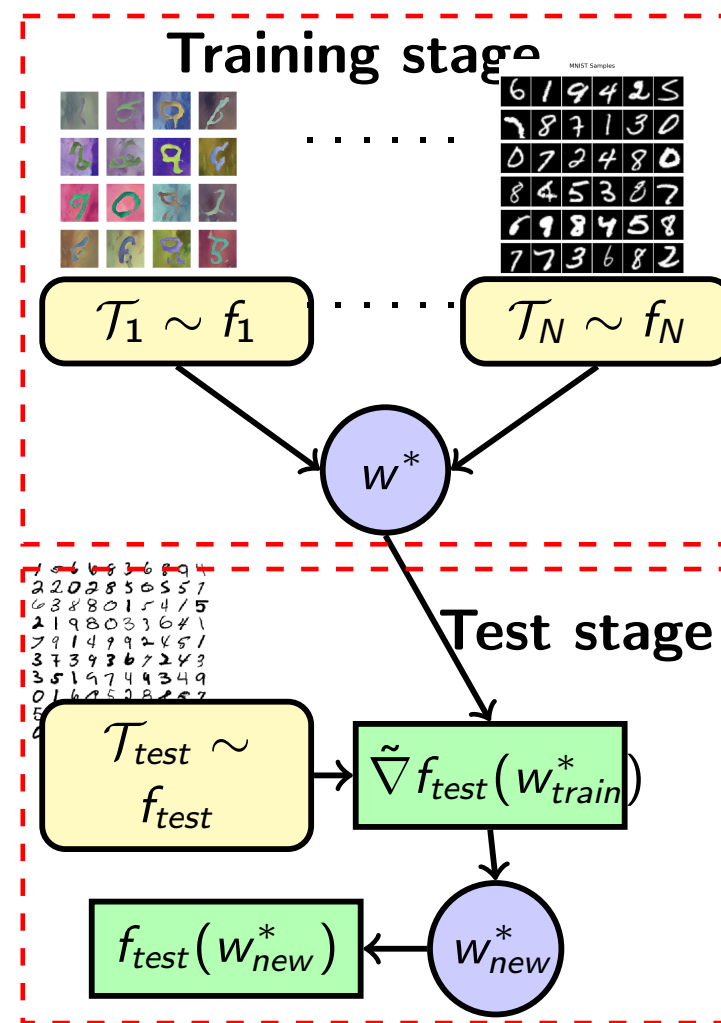
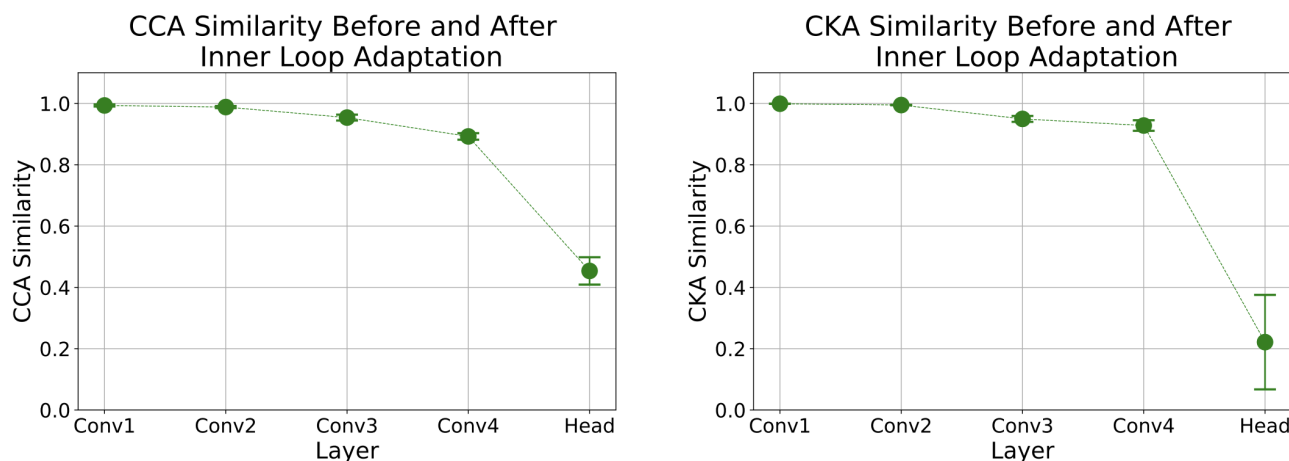


Image Credits: <https://bit.ly/392pda9>,
<https://bit.ly/3EEIElq>

Empirical Observations of MAML

- 1 MAML learns models that can **quickly solve new tasks** [FAL17, AES19]
 - ▶ in image classification, sinusoid regression, reinforcement learning.
- 2 MAML seems to be learning a representation **shared across tasks** [RRBV20]
 - ▶ even though it is not designed for representation learning!



The representation learned by MAML does not change significantly when adapted to each task.
Figure credit: [RRBV20]

Part 2b: MAML in a Multi-Task Linear Setting

Meta-learning

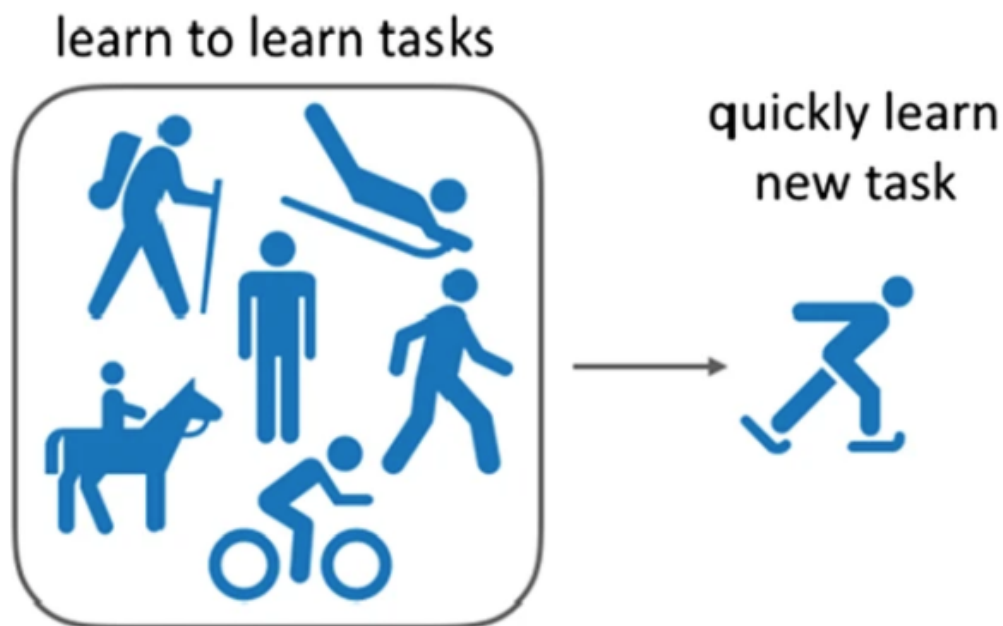


Image credit: <https://meta-world.github.io>, [HRJ21]

Representation learning in multi-task linear regression (1/2)

- Example: consider multi-task linear regression.
- Task i has ground-truth solution $\theta_{*,i} \in \mathbb{R}^d$:

$$y_i \sim \theta_{*,i}^\top \mathbf{x}_i + z_i$$

where $\mathbf{x}_i$ is a random feature vector and $z_i \in \mathbb{R}$ is random, mean-zero noise.

- Solving each task individually (i.e. finding a $\theta_i \approx \theta_{*,i}$ for each i) would require $\Omega(d)$ samples per task.

Question

Can we do better using shared information across tasks?

Representation learning in multi-task linear regression (2/2)

- Now suppose the $\theta_{*,i}$ lie in a shared k -dimensional subspace, $k \ll d$
- Let the columns of $\mathbf{B}_* \in \mathbb{R}^{d \times k}$ span this subspace, that is, for all tasks there exists $\mathbf{w}_{*,i} \in \mathbb{R}^k$ such that

$$\theta_{*,i} = \mathbf{B}_* \mathbf{w}_{*,i}$$

- ▶ $\mathbf{B}_*$ is the “ground-truth” representation
- If we know $\text{col}(\mathbf{B}_*)$, we can solve new tasks with only $O(k)$ samples

Benefit of Representation Learning

$O(k)$ sample complexity much smaller than $\Omega(d)$

MAML and ANIL for multi-task linear regression (1/2)

- Loss function for task i at round t :

$$f_i(\mathbf{B}, \mathbf{w}) := \frac{1}{2} \mathbb{E}_{\mathbf{x}_i, y_i} [(\langle \mathbf{B}\mathbf{w}, \mathbf{x}_i \rangle - y_i)^2]$$

- Large population of tasks (all of which lie in the subspace $\mathbf{B}_*$)
- At each time t , we sample n distinct tasks (may differ across time)
- For each task, we collect $m_{in} + m_{out}$ data samples (approximates the expectation in the population loss in the **inner** and **outer** loops)

Algorithm (MAML)

- (Outer loop) For $t = 1, \dots, T$:

- Select n tasks satisfying diversity condition ($\text{span}(\{\mathbf{w}_{*,i}\}_{i \in [n]}) = \mathbb{R}^k$)
- (Inner loop) For $i = 1, \dots, n$:
 - Adapt: $\mathbf{w}_{t,i} = \mathbf{w}_t - \alpha \nabla_{\mathbf{w}} f_i(\mathbf{B}_t, \mathbf{w}_t)$, $\mathbf{B}_{t,i} = \mathbf{B}_t - \alpha \nabla_{\mathbf{B}} f_i(\mathbf{B}_t, \mathbf{w}_t)$

$$\begin{bmatrix} \mathbf{w}_{t+1} \\ \bar{\mathbf{B}}_{t+1} \end{bmatrix} = \begin{bmatrix} \mathbf{w}_t \\ \bar{\mathbf{B}}_t \end{bmatrix} - \frac{\beta}{n} \sum_{i=1}^n (\mathbf{I} - \alpha \nabla_{\mathbf{w}, \bar{\mathbf{B}}}^2 f_i(\mathbf{B}_t, \mathbf{w}_t)) \begin{bmatrix} \nabla_{\mathbf{w}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \\ \nabla_{\mathbf{B}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \end{bmatrix}$$

where $\bar{\mathbf{B}}$ is the column-wise vectorization of $\mathbf{B}$.

Change the Loss Function

$$\theta_{train}^* \in \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n f_i(\theta - \alpha \nabla f_i(\theta; (X_i, y_i)))$$

MAML and ANIL for multi-task linear regression (2/2)

- Loss function for task i at round t :

$$f_i(\mathbf{B}, \mathbf{w}) := \frac{1}{2} \mathbb{E}_{\mathbf{x}_i, y_i} [(\langle \mathbf{B}\mathbf{w}, \mathbf{x}_i \rangle - y_i)^2]$$

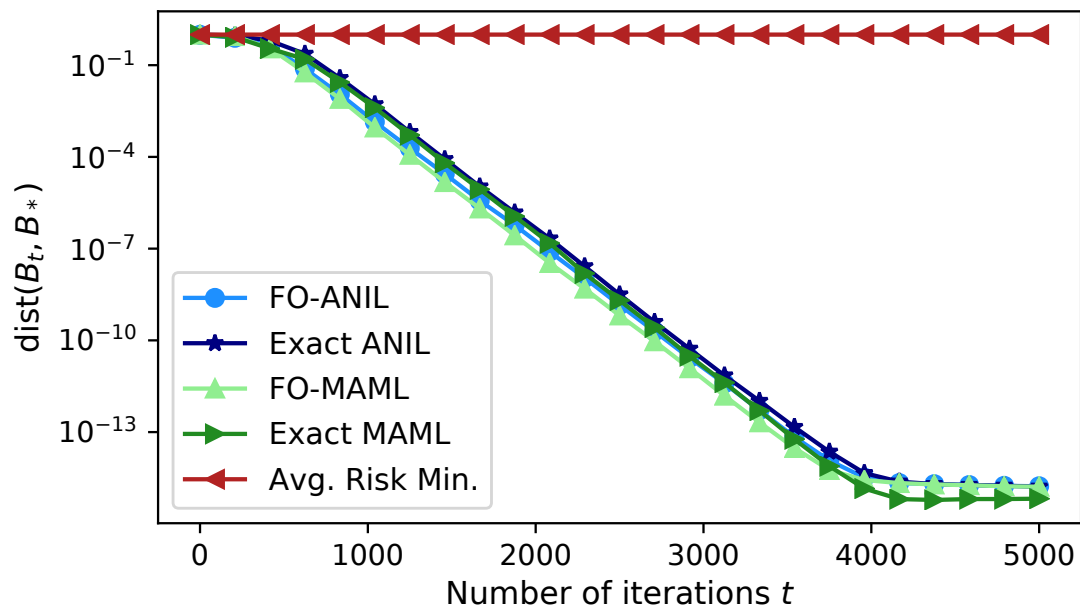
- At each time t , we sample n distinct tasks (may differ across time)
- We assume access to population losses in description for simplicity

Algorithm

- (Outer loop) For $t = 1, \dots, T$:
 - Select n tasks satisfying diversity condition ($\text{span}(\{\mathbf{w}_{*,i}\}_{i \in [n]}) = \mathbb{R}^k$)
 - (Inner loop) For $i = 1, \dots, n$:
 - ★ - Adapt head: $\mathbf{w}_{t,i} = \mathbf{w}_t - \alpha \nabla_{\mathbf{w}} f_i(\mathbf{B}_t, \mathbf{w}_t)$.
 - ★ - If MAML: Adapt rep: $\mathbf{B}_{t,i} = \mathbf{B}_t - \alpha \nabla_{\mathbf{B}} f_i(\mathbf{B}_t, \mathbf{w}_t)$
If ANIL: Do not adapt rep: $\mathbf{B}_{t,i} = \mathbf{B}_t$
- $\begin{bmatrix} \mathbf{w}_{t+1} \\ \bar{\mathbf{B}}_{t+1} \end{bmatrix} = \begin{bmatrix} \mathbf{w}_t \\ \bar{\mathbf{B}}_t \end{bmatrix} - \frac{\beta}{n} \sum_{i=1}^n \mathbf{H}_{t,i,\text{Alg}}(\mathbf{B}_t, \mathbf{w}_t) \begin{bmatrix} \nabla_{\mathbf{w}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \\ \nabla_{\mathbf{B}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \end{bmatrix}$ where $\mathbf{H}_{t,i,\text{Alg}}(\mathbf{B}_t, \mathbf{w}_t)$ is a Hessian that differs between MAML and ANIL.

MAML: Evidence of representation learning in linear setting

- We consider four meta-learning algorithms:
 - ▶ MAML,
 - ▶ ANIL [RRBV19], a close relative of MAML, and
 - ▶ their first-order approximations (FO-MAML and FO-ANIL).



For multi-task linear regression with population losses, the **meta-learning approaches** learn $\text{col}(\mathbf{B}_*)$, while **ERM does not**.

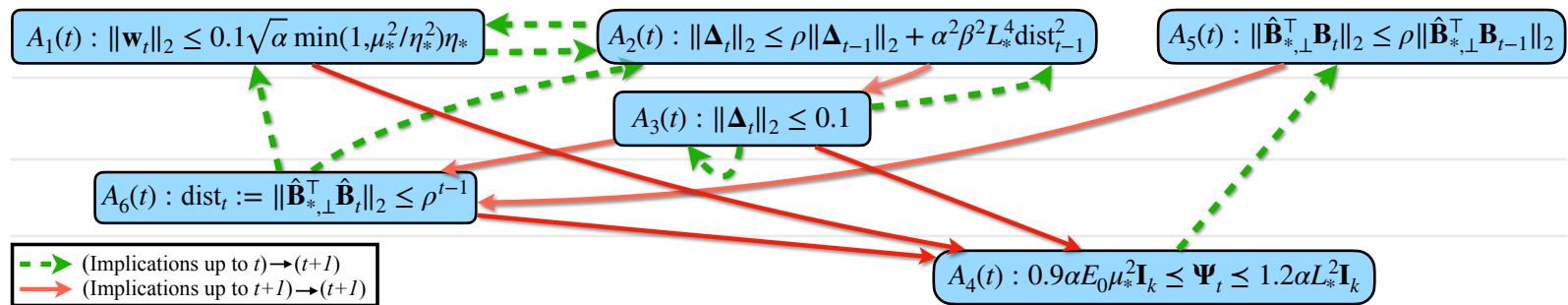
How can we explain this?

- Despite the empirical evidence that MAML and ANIL learn high-quality representations, it is difficult to understand *how* they are able to do so.
 - ▶ MAML is *model-agnostic* – not designed for representation learning!
- Existing works have yet to solve this mystery.
 - ▶ **No existing theory** for whether and how MAML or ANIL learn good representations, in any setting.
 - ▶ MAML and ANIL involve inner and outer loops → **difficult to analyze**.

Takeaway

We must better understand MAML and ANIL dynamics to explain why they learn representations.

Part 2c: Representation Learning with MAML/ANIL



Main Results

- We consider the multi-task linear regression setting

Main Results

- *Under standard assumptions, MAML, ANIL and their first-order analogues recover $\text{col}(\mathbf{B}_*)$ exponentially fast when run on the task population losses.*
- *ANIL and FO-ANIL require $m = \Omega((\frac{d}{n} + 1)k^3) \ll d$ samples per task to learn the ground-truth subspace.*
- *The key is that MAML and ANIL's adaptation of the head harnesses **task diversity** to improve the representation in all directions.*
- **First results** showing that MAML and ANIL provably learn effective representations!

Proof Intuition

$$\text{Adapt: } \mathbf{w}_{t,i} = \mathbf{w}_t - \alpha \nabla_{\mathbf{w}} f_i(\mathbf{B}_t, \mathbf{w}_t), \mathbf{B}_{t,i} = \mathbf{B}_t - \alpha \nabla_{\mathbf{B}} f_i(\mathbf{B}_t, \mathbf{w}_t)$$
$$\begin{bmatrix} \mathbf{w}_{t+1} \\ \bar{\mathbf{B}}_{t+1} \end{bmatrix} = \begin{bmatrix} \mathbf{w}_t \\ \bar{\mathbf{B}}_t \end{bmatrix} - \frac{\beta}{n} \sum_{i=1}^n (\mathbf{I} - \alpha \nabla_{\mathbf{w}, \bar{\mathbf{B}}}^2 f_i(\mathbf{B}_t, \mathbf{w}_t)) \begin{bmatrix} \nabla_{\mathbf{w}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \\ \nabla_{\mathbf{B}} f_i(\mathbf{B}_{t,i}, \mathbf{w}_{t,i}) \end{bmatrix}$$

- For FO-ANIL, we have

$$\mathbf{B}_{t+1} = \mathbf{B}_t \underbrace{\left(\mathbf{I}_k - \frac{\beta}{n} \sum_{i=1}^n \mathbf{w}_{t,i} \mathbf{w}_{t,i}^\top \right)}_{\text{prior weight}} + \mathbf{B}_* \underbrace{\frac{\beta}{n} \sum_{i=1}^n \mathbf{w}_{*,t,i} \mathbf{w}_{t,i}^\top}_{\text{signal weight}}$$

- Suppose $\frac{1}{n} \sum_{i=1}^n \mathbf{w}_{t,i} \mathbf{w}_{t,i}^\top$ is full rank (i.e., the $\mathbf{w}_{t,i}$'s are diverse), then:

Key observation

Prior weight reduces energy from $\mathbf{B}_t$, and **signal weight** boosts energy from $\mathbf{B}_*$ in all directions.

$\implies$ **Head adaptation and task diversity are critical!**

- The **more diverse** the **adapted** $\mathbf{w}_{t,i}$'s (i.e. smaller condition number of $\frac{1}{n} \sum_{i=1}^n \mathbf{w}_{t,i} \mathbf{w}_{t,i}^\top$), the **faster convergence rate**.

Comparison to Average Risk Minimization (1/2)

$$\text{ERM: } \min_{\mathbf{B}, \mathbf{w}} \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{B}, \mathbf{w})$$

- In this case we can show:

$$\mathbf{B}_{t+1} = \mathbf{B}_t \left(\underbrace{\mathbf{I}_k - \beta \mathbf{w}_t \mathbf{w}_t^\top}_{\text{prior weight}} \right) + \mathbf{B}_* \underbrace{\frac{\beta}{n} \sum_{i=1}^n \mathbf{w}_{*,t,i} \mathbf{w}_t^\top}_{\text{signal weight}}$$

- The **prior weight** is rank $k - 1$, while the **signal weight** is only rank 1.

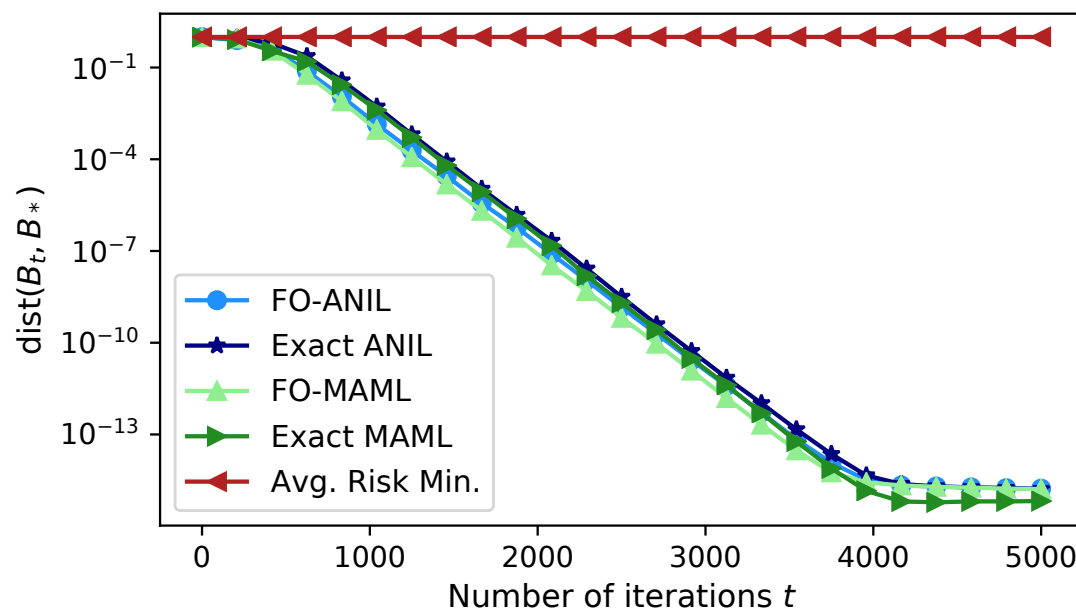
Key observation

The representation can only move closer to $\text{col}(\mathbf{B}_*)$ in **one** direction on each iteration $\rightarrow$ not clear that it can eventually reach $\text{col}(\mathbf{B}_*)$.

Comparison to Average Risk Minimization (2/2)

$$\text{ARM: } \min_{\mathbf{B}, \mathbf{w}} \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{B}, \mathbf{w})$$

- Empirically, Average Risk Minimization fails to learn $\text{col}(\mathbf{B}_*)$.



Challenges of Proving Representation Learning (1/2)

- Need to show that the $\mathbf{w}_{t,i}$'s are sufficiently diverse, i.e. evenly spread across $\mathbb{R}^k$.
- We can show:

$$\begin{aligned}\mathbf{w}_{t,i} &= \mathbf{w}_t - \alpha \nabla f_i(\mathbf{B}_t, \mathbf{w}_t) \\ &= \underbrace{(\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t) \mathbf{w}_t}_{\text{shared for all } i} + \underbrace{\alpha \mathbf{B}_t^\top \mathbf{B}_* \mathbf{w}_{*,t,i}}_{\text{unique for each } i}\end{aligned}$$

- We must show the unique part of $\mathbf{w}_{t,i}$ dominates the shared part.

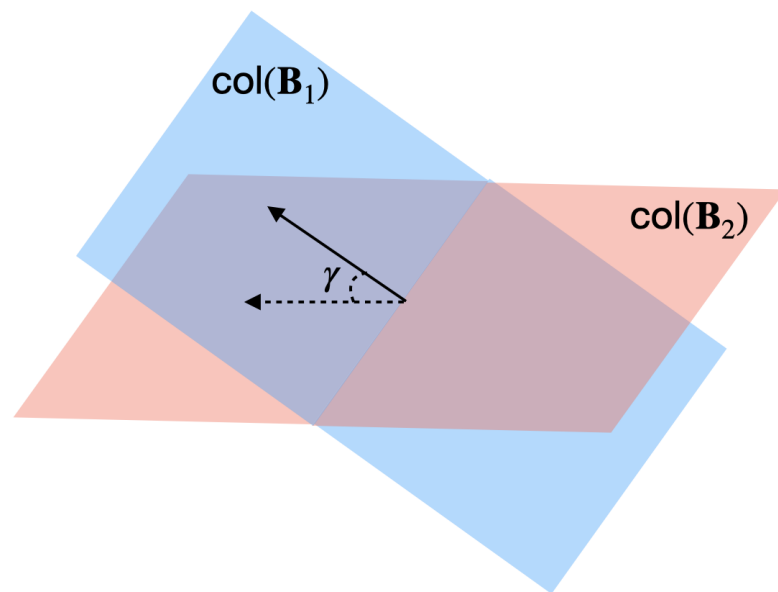
Challenges of Proving Representation Learning (2/2)

$$\mathbf{w}_{t,i} = \underbrace{(\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t) \mathbf{w}_t}_{\text{shared for all } i} + \underbrace{\alpha \mathbf{B}_t^\top \mathbf{B}_* \mathbf{w}_{*,t,i}}_{\text{unique for each } i}$$

- In order to show the unique part dominates, we must show three things hold for all t :
 - 1 $\|\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t\|_2$ is small
 - 2 $\|\mathbf{w}_t\|_2$ is small
 - 3 $\sigma_{\min}(\mathbf{B}_t^\top \mathbf{B}_*)$ is lower bounded $\iff \text{dist}(\mathbf{B}_t, \mathbf{B}_*) < 1 - c$ for a positive constant c
- Difficult because the algorithms lack explicit regularization and a normalization step.

Principal Angle Distance

- We use the **principal angle distance** to measure the distance between representations.



- Formally,

$$\text{dist}(\mathbf{B}_1, \mathbf{B}_2) := \|\hat{\mathbf{B}}_{1,\perp}^\top \hat{\mathbf{B}}_2\|_2,$$

where $\hat{\mathbf{B}}_{1,\perp}$ and $\hat{\mathbf{B}}_2$ are orthonormal matrices s.t.
 $\text{col}(\hat{\mathbf{B}}_{1,\perp}) = \text{col}(\mathbf{B}_1)^\perp$ and $\text{col}(\hat{\mathbf{B}}_2) = \text{col}(\mathbf{B}_2)$.

Population case result

(Informal) [Collins-Mokhtari-Oh-S]

Suppose there are $m = \infty$ samples/task, the ground-truth heads satisfy $\mu_*^2 \mathbf{I}_k \preceq \frac{1}{n} \sum_{i=1}^n \mathbf{w}_{*,t,i} \mathbf{w}_{*,t,i}^\top \preceq L_*^2 \mathbf{I}_k$, and the step sizes α, β are sufficiently small. Then after T iterations, ANIL, FO-ANIL, MAML, and FO-MAML learn a representation $\mathbf{B}_T$ that satisfies:

$$\text{dist}(\mathbf{B}_T, \mathbf{B}_*) \leq (1 - \Omega(\beta \alpha \mu_*^2))^{T-1}$$

as long as:

- ANIL, FO-ANIL: $\text{dist}(\mathbf{B}_0, \mathbf{B}_*) \leq c$ for a constant c .
- MAML: $\text{dist}(\mathbf{B}_0, \mathbf{B}_*) = O((L_*/\mu_*)^{-0.75})$.
- FO-MAML: $\text{dist}(\mathbf{B}_0, \mathbf{B}_*) = O((L_*/\mu_*)^{-1})$ and $\|\frac{1}{n} \sum_{i=1}^n \mathbf{w}_{*,t,i}\|_2 = O((L_*/\mu_*)^{-1.5})$.

- We also show finite-sample results in the paper.

Proof sketch - FO-ANIL (1/5)

$$\mathbf{w}_{t,i} = \underbrace{(\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t) \mathbf{w}_t}_{\text{shared for all } i} + \underbrace{\alpha \mathbf{B}_t^\top \mathbf{B}_* \mathbf{w}_{*,t,i}}_{\text{unique for each } i}$$

- Recall that we need to show the $\mathbf{w}_{t,i}$'s are diverse, which requires:
 - 1 $\|\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t\|_2$ is small
 - 2 $\|\mathbf{w}_t\|_2$ is small
 - 3 $\sigma_{\min}(\mathbf{B}_t^\top \mathbf{B}_*)$ is lower bounded $\iff \text{dist}(\mathbf{B}_t, \mathbf{B}_*) < 1 - c$ for a positive constant c
- Leads to an intricate 6-way induction....

Proof sketch - FO-ANIL (2/5)

→ "well conditioned representation"

$$\mathbf{w}_{t,i} = \underbrace{(\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t) \mathbf{w}_t}_{\text{shared for all } i} + \underbrace{\alpha \mathbf{B}_t^\top \mathbf{B}_* \mathbf{w}_{*,t,i}}_{\text{unique for each } i}$$

- Define $\Delta_t := \mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t$, $\Psi_t := \frac{1}{n} \sum_{i=1}^n \mathbf{w}_{t,i} \mathbf{w}_{t,i}^\top$, and $\rho := 1 - \Omega(\beta\alpha)$.
→ "diverse heads"

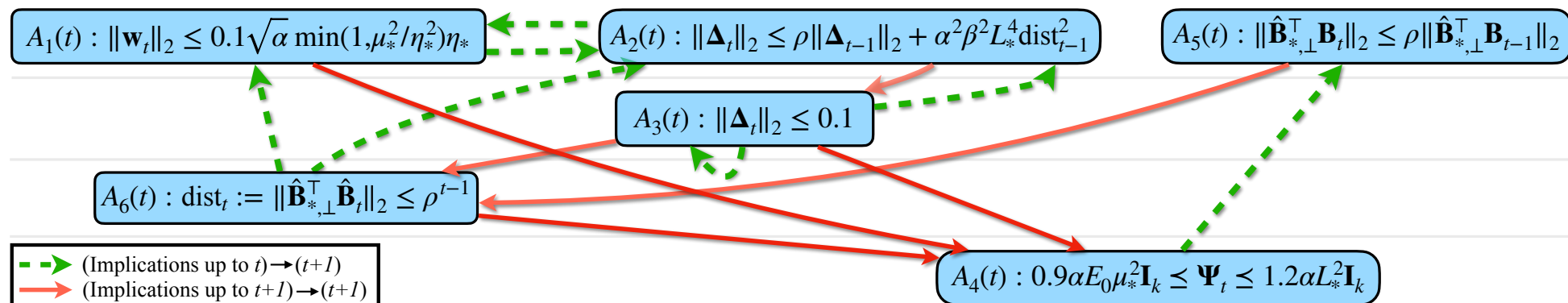
- Inductive hypotheses:

- ① $A_1(t) := \{\|\mathbf{w}_t\|_2 = O(\sqrt{\alpha})\}$ - global head stays small
- ② $A_2(t) := \{\|\Delta_t\|_2 = \rho \|\Delta_{t-1}\|_2 + O(\beta^2 \alpha^2 \text{dist}_{t-1}^2)\}$ - $\mathbf{B}_t$ gets closer to orthogonal as dist_t decreases
- ③ $A_3(t) := \{\|\Delta_t\|_2 = O(1)\}$ - $\mathbf{B}_t$ stays uniformly close to orthogonal
- ④ $A_4(t) := \{\kappa(\Psi_t) = O(1)\}$ - adapted heads $\mathbf{w}_{t,i}$ are diverse
- ⑤ $A_5(t) := \{\|\mathbf{B}_{*,\perp}^\top \mathbf{B}_t\|_2 = \rho \|\mathbf{B}_{*,\perp}^\top \mathbf{B}_{t-1}\|_2\}$ - energy of $\mathbf{B}_t$ in perpendicular space to $\text{col}(\mathbf{B}_*)$ is contracting
- ⑥ $A_6(t) := \{\text{dist}_t \leq \rho^{t-1}\}$ - PA distance of $\mathbf{B}_t$ to $\mathbf{B}_*$ is linearly decreasing

Proof sketch - FO-ANIL (3/5)

- Inductive logic:

$$\mathbf{B}_{t+1} = \mathbf{B}_t \underbrace{\left(\mathbf{I}_k - \frac{\beta}{n} \sum_{i=1}^n \mathbf{w}_{t,i} \mathbf{w}_{t,i}^\top \right)}_{\text{prior weight}} + \mathbf{B}_* \underbrace{\frac{\beta}{n} \sum_{i=1}^n \mathbf{w}_{*,t,i} \mathbf{w}_{t,i}^\top}_{\text{signal weight}}$$



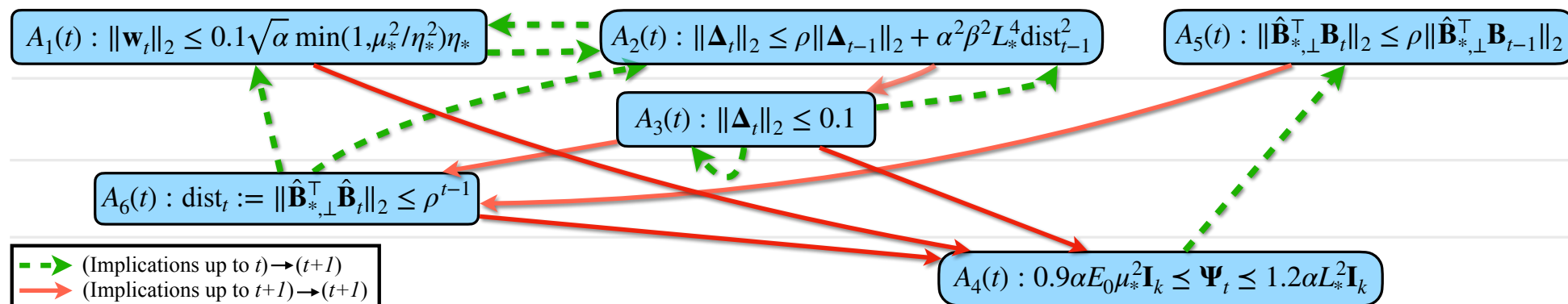
Notable implications (1/3):

- $A_4(t) \implies A_5(t+1) \xrightarrow{A_3(t+1)} A_6(t+1)$: Adapted head diversity and well-conditioned $\mathbf{B}_{t+1}$ implies dist_{t+1} is linearly converging.
 - Proof.* Use expression for $\mathbf{B}_{t+1}$ discussed earlier.

Proof sketch - FO-ANIL (4/5)

$$\mathbf{w}_{t,i} = \underbrace{(\mathbf{I}_k - \alpha \mathbf{B}_t^\top \mathbf{B}_t) \mathbf{w}_t}_{\text{shared for all } i} + \underbrace{\alpha \mathbf{B}_t^\top \mathbf{B}_* \mathbf{w}_{*,t,i}}_{\text{unique for each } i}$$

- Inductive logic:

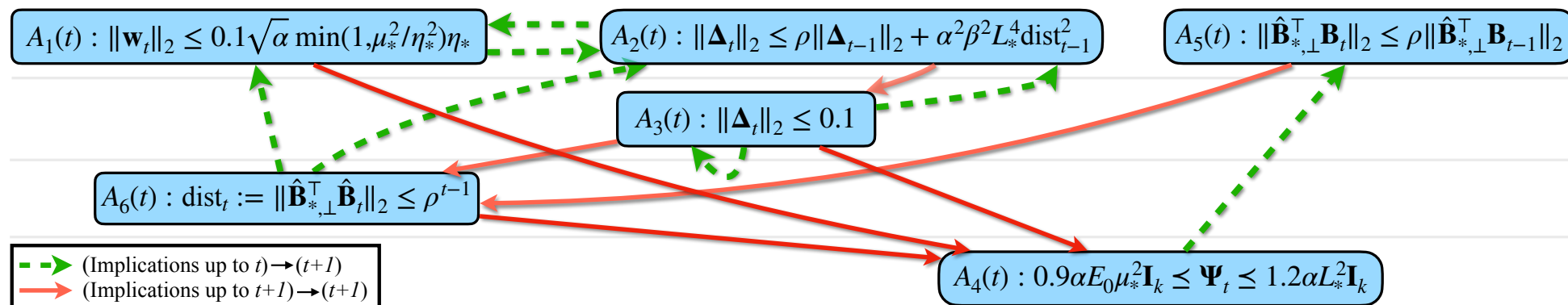


Notable implications (2/3):

- $A_1(t+1), A_3(t+1), A_6(t+1) \implies A_4(t+1)$: Small $\|\mathbf{w}_t\|_2$, $\|\Delta_t\|_2$, and dist_t implies adapted heads are diverse.
 - *Proof.* Use expression for $\mathbf{w}_{t,i}$ discussed earlier.

Proof sketch - FO-ANIL (5/5)

- Inductive logic:

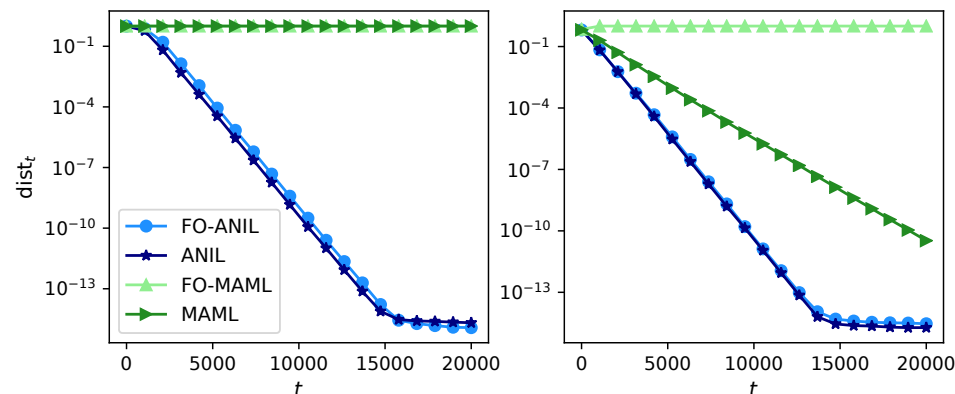


Notable implications (3/3):

- $A_2(t) + A_6(t) \implies A_1(t+1)$: Bounds on $\|\Delta_t\|_2$ and dist_t imply $\|\mathbf{w}_{t+1}\|_2$ stays small.
 - Proof.* $A_2(t)$ and the linear convergence of dist_t implies $\|\Delta_t\|_2$ eventually linearly converges to zero. We can show $\|\mathbf{w}_{t+1}\|_2 \leq \|\mathbf{w}_t\|_2 + O(\|\Delta_t\|_2)$, which implies $|\|\mathbf{w}_{t+1}\|_2 - \|\mathbf{w}_t\|_2|$ eventually linearly converges to zero.

MAML vs. ANIL

- Recall that our result requires
 - stronger initialization for MAML and FO-MAML than for ANIL and FO-ANIL, and
 - for FO-MAML, the mean of $\mathbf{w}_{*,t,i}$'s must be ≈ 0 .
- Here we show these conditions are tight:



(Left) Random initialization and (Right) structured initialization. In both cases, the mean of the ground-truth heads is far from zero, explaining why FO-MAML fails. MAML succeeds only with structured initialization.

⇒ MAML/FO-MAML's inner loop update of the representation can inhibit representation learning.

Discussion

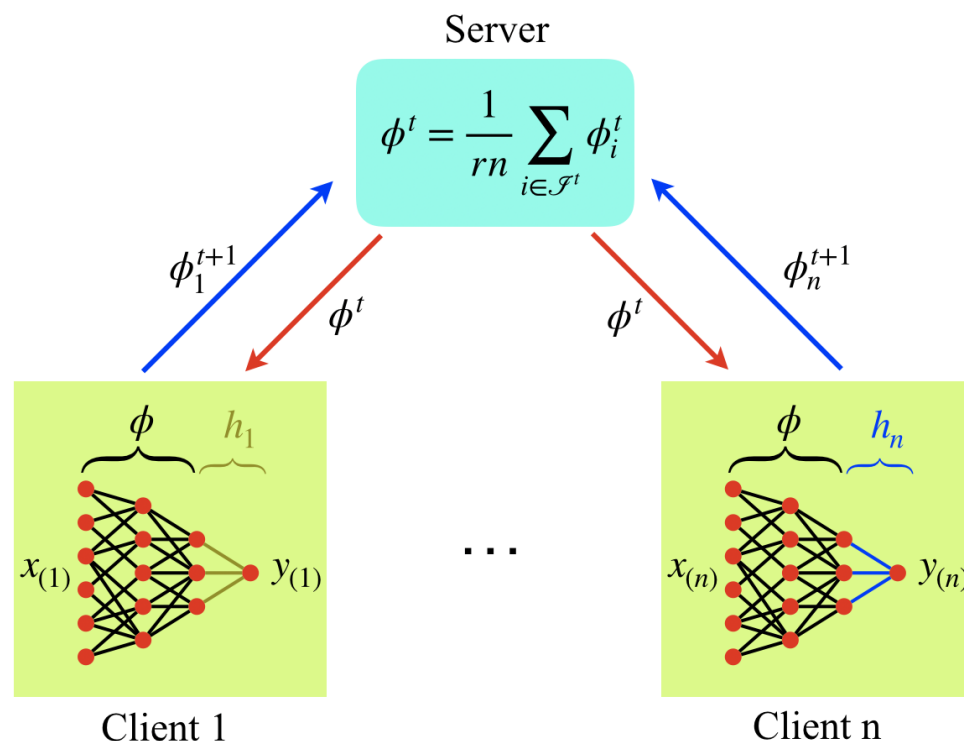
- We have obtained the **first results** showing that ANIL and MAML learn effective representations in any setting.¹
- Inner loop adaptation of the head is **key** to MAML and ANIL's ability to learn representations.
- Inner loop adaptation of the representation **restricts representation learning**.
- Suggests the need for algorithms that augment and exploit task diversity to learn representations.

¹L. Collins, A. Mokhtari, S. Oh, S. Shakkottai. MAML and ANIL Provably Learn Representations, ICML 2022: <https://arxiv.org/abs/2202.03483>

References

- [FAL17] Chelsea Finn, Pieter Abbeel, Sergey Levine. Model-Agnostic Meta-Learning for Fast Adaptation of Neural Networks, *International Conference on Machine Learning*, 2017.
- [AES19] Antreas Antoniou, Harrison Edwards, Amos Storkey. How to Train Your MAML, *International Conference on Learning Representations*, 2019.
- [RRBV19] Aniruddh Raghu, Maithra Raghu, Samy Bengio, Oriol Vinyals. Rapid Learning or Feature Reuse? Towards Understanding the Effectiveness of MAML, *International Conference on Learning Representations*, 2020.
- [HRJ21] Mike Huisman, Jan N. van Rijn, Aske Plaat. A Survey of deep Meta-Learning, *Artificial Intelligence Review* Volume 54, pages 4483–4541, 2021.

Part 3: Changing the way we train – the FedAvg approach



Recap: Average Risk Minimization (ARM)

- **Step 1:** Train a **single model** by minimizing the average loss

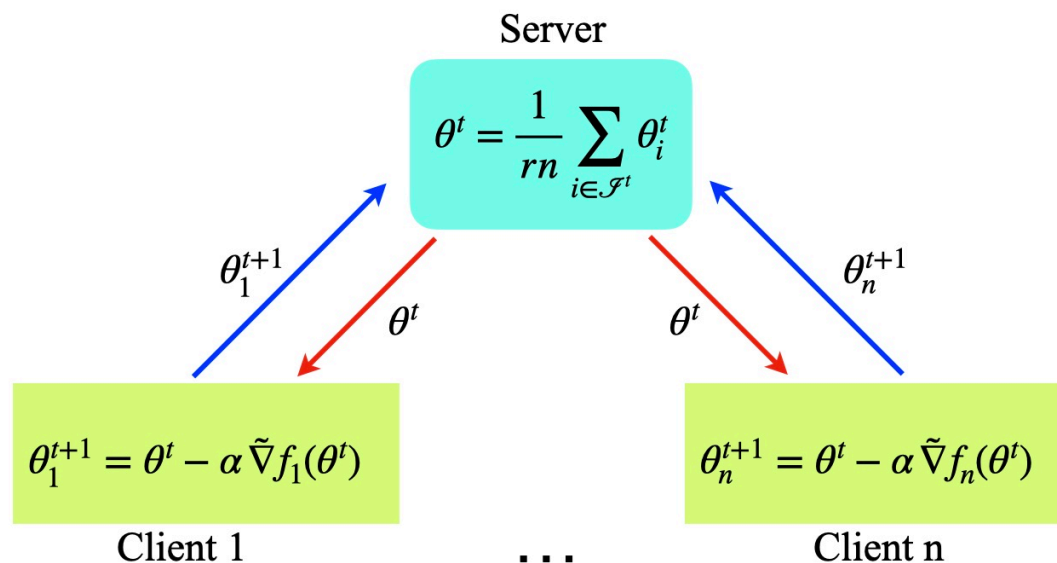
$$\theta^* \in \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n f_i(\theta; \mathcal{D}_i)$$

- **Step 2:** **Fine tune** the obtained model θ^* to new task / deployment environment



A Distributed Implementation: DSGD

- Popular approach: Distributed Stochastic Gradient Descent (DSGD)



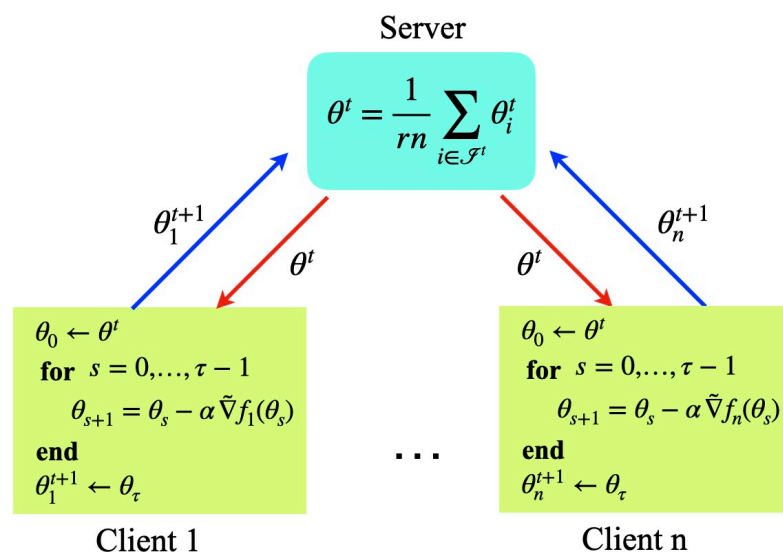
- Effective update $\theta^{t+1} = \theta^t - \frac{\alpha}{rn} \sum_{i \in \mathcal{I}^t} \tilde{\nabla} f_i(\theta^t)$
 $\Rightarrow$ running SGD on the global loss

Same Loss as ARM – Only change is a Distributed Implementation

$$\theta^* \in \operatorname{argmin}_{\theta} \frac{1}{n} \sum_{i=1}^n f_i(\theta; \mathcal{D}_i)$$

Federated Learning: A Communication-Efficient Approach

- Federated learning is similar to DSGD
 - ▶ Except it allows for **multiple local updates**
- FedAvg: the most common approach!



- **Advantage:** Number of communication rounds $\ll$ number of updates
 - ▶ More communication-efficient than distributed SGD!
- **Issue:** Drifting effect with data heterogeneity (aka multiple tasks)
 - ▶ The local grads ∇f_i are not aligned with the global direction $\sum_{i=1}^n \nabla f_i$

The Effect of Local Updates

From an **optimization point of view**:

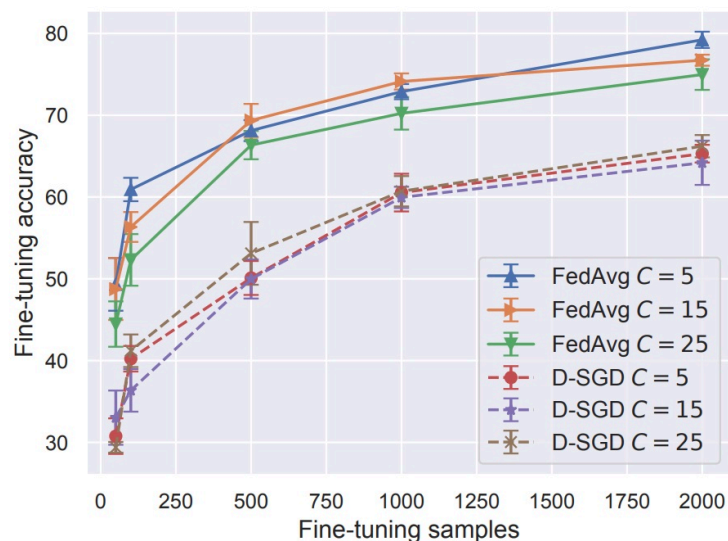
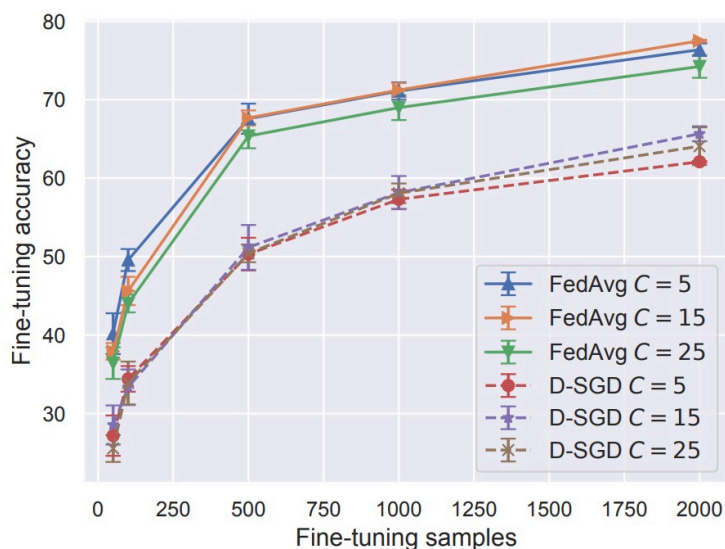
- Due to local drifting, FedAvg may not solve global objective in the data heterogeneous setting
 - ▶ [Wang & Joshi, 2019], [Li et. al., 2020], [Charles & Konecny, 2021],
- There are several other sophisticated methods to control/remove the local drift issue in FedAvg
 - ▶ SCAFFOLD: [Karimireddy et. al., 2019]
 - ▶ VRL-SGD: [Liang et. al., 2019]
 - ▶ FEDGATE: [Haddadpour et. al., 2020]
 - ▶ FedNova: [Wang et. al., 2020]
 - ▶

Our Takeaway

Implicitly, the new implementation has changed the objective

Local updates in FedAvg are useful!

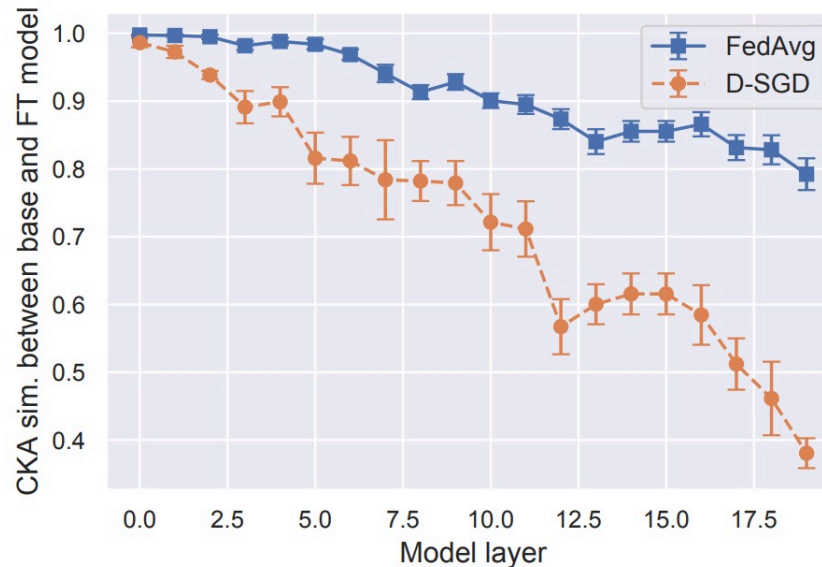
- The local updates are in fact good for **generalization**!



- Left plot:** Models trained on 80 classes from CIFAR-100 (with C classes/client) and fine-tuned on new clients from 20 new classes from CIFAR-100
- Right plot:** Models trained on CIFAR-100 (with C classes/client) and fine-tuned on new clients from CIFAR-10
- $T\tau = 125000$ for both. (FedAvg $\tau = 50$, $T = 2500$, DSGD $\tau = 1$, $T = 125000$)

FedAvg and Representation Learning

- The early layers of FedAvg's pre-trained model (corresponding to the representation) change much less than those of D-SGD



- Local updates enable learning the common representation among the tasks!

Open Problems

- *Does FedAvg provably learn representations of heterogeneous tasks?*
- *Are the local updates essential for learning representations?*

Setup: (Same as before) Multi-Task Linear Regression

- Multi-task linear regression.
- Task i has ground-truth solution $\theta_{*,i} \in \mathbb{R}^d$:

$$y_i \sim \theta_{*,i}^\top \mathbf{x}_i + z_i$$

► $\mathbf{x}_i$ is a random feature vector, $z_i \in \mathbb{R}$ is random, mean-zero noise

- Now suppose the $\theta_{*,i}$ lie in a shared k -dimensional subspace, $k \ll d$
- Let the columns of $\mathbf{B}_* \in \mathbb{R}^{d \times k}$ span this subspace, that is, for all tasks there exists $\mathbf{w}_{*,i} \in \mathbb{R}^k$ such that

$$\theta_{*,i} = \mathbf{B}_* \mathbf{w}_{*,i}$$

Task Diversity: The concatenation of $\mathbf{w}_{*,1}, \dots, \mathbf{w}_{*,n}$ spans $\mathbb{R}^k$

Main Results (FedAvg Learns Representation)

Theorem (informal) [Collins-Hassani-Mokhtari-S]

If the number of local updates satisfies $\tau \geq 2$, FedAvg recovers $\text{col}(\mathbf{B}^)$ exponentially fast when run on the task population losses.*

- The key insight is that FedAvg local updates harnesses **task diversity** to improve the representation in all directions.

$$\mathbf{B}_{t+1} = \mathbf{B}_t \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \prod_{s=0}^{\tau-1} \left(\mathbf{I}_k - \alpha \mathbf{w}_{t,i,s} \mathbf{w}_{t,i,s}^\top \right) \right)}_{\text{prior weight}} + \underbrace{\mathbf{B}_* \left(\frac{\alpha}{n} \sum_{i=1}^n \mathbf{w}_{*,i} \sum_{s=0}^{\tau-1} \mathbf{w}_{t,i,s}^\top \prod_{r=s+1}^{\tau-1} \left(\mathbf{I}_k - \alpha \sum_{i=1}^n \mathbf{w}_{t,i,r} \mathbf{w}_{t,i,r}^\top \right) \right)}_{\text{signal weight}}$$

- **Prior weight** reduces energy from $\mathbf{B}_t$, and **signal weight** boosts energy from $\mathbf{B}_*$ in all directions

Main Results (FedAvg Learns Representation)

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$$\mathbf{B}_{t+1} \approx \mathbf{B}_t \underbrace{\left(\mathbf{I}_k - \frac{\alpha}{n} \sum_{i=1}^n \sum_{s=0}^{\tau-1} \mathbf{w}_{t,i,s} \mathbf{w}_{t,i,s}^\top \right)}_{\text{prior weight}} + \mathbf{B}_* \underbrace{\left(\frac{\alpha}{n} \sum_{i=1}^n \sum_{s=0}^{\tau-1} \mathbf{w}_{*,i} \mathbf{w}_{t,i,s}^\top \right)}_{\text{signal weight}}$$

- **Prior weight** reduces energy from $\mathbf{B}_t$, and **signal weight** boosts energy from $\mathbf{B}_*$ in all directions
 - ▶ **Local updates** and **task diversity** are critical!

Main Results (DGD May Not Learn Representation)

- Now suppose $\tau = 1$ and we have the update of DGD

$$\mathbf{B}_{t+1} = \mathbf{B}_t \underbrace{\left(\mathbf{I}_k - \alpha \mathbf{w}_t \mathbf{w}_t^\top \right)}_{\text{prior weight}} + \alpha \mathbf{B}_* \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \mathbf{w}_{*,i} \right) \mathbf{w}_t^\top}_{\text{signal weight}}$$

- The **prior weight** is rank $k - 1$, while the **signal weight** is only rank 1.
- The representation can only move closer to $\text{col}(\mathbf{B}_*)$ in **one** direction on each iteration
 - not clear that it can eventually reach $\text{col}(\mathbf{B}_*)$

Theorem (informal) [Collins-Hassani-Mokhtari-S]

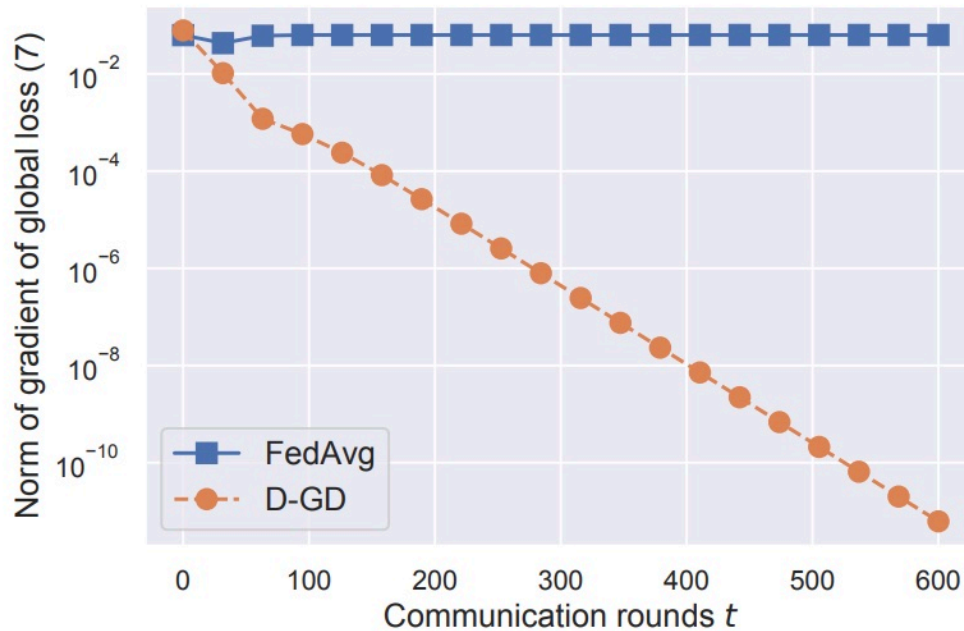
For any $\delta \in (0., 0.5]$, α , T , $\{\mathbf{w}_{*,i}\}$ and full rank $\mathbf{B}_0$, there exists a $\mathbf{B}_*$ whose column space is δ -close to $\text{col}(\mathbf{B}_0)$, i.e., $\text{dist}(\mathbf{B}_0, \mathbf{B}_*) = \delta$, while its distance from the representation learned by DGD is at least 0.7δ , i.e., $\text{dist}(\mathbf{B}_T^{\text{DGD}}, \mathbf{B}_*) > 0.7\delta$.

- DGD cannot guarantee to recover the ground-truth representation.
⇒ Local updates are necessary!

Important Observations and Conclusion (1/2)

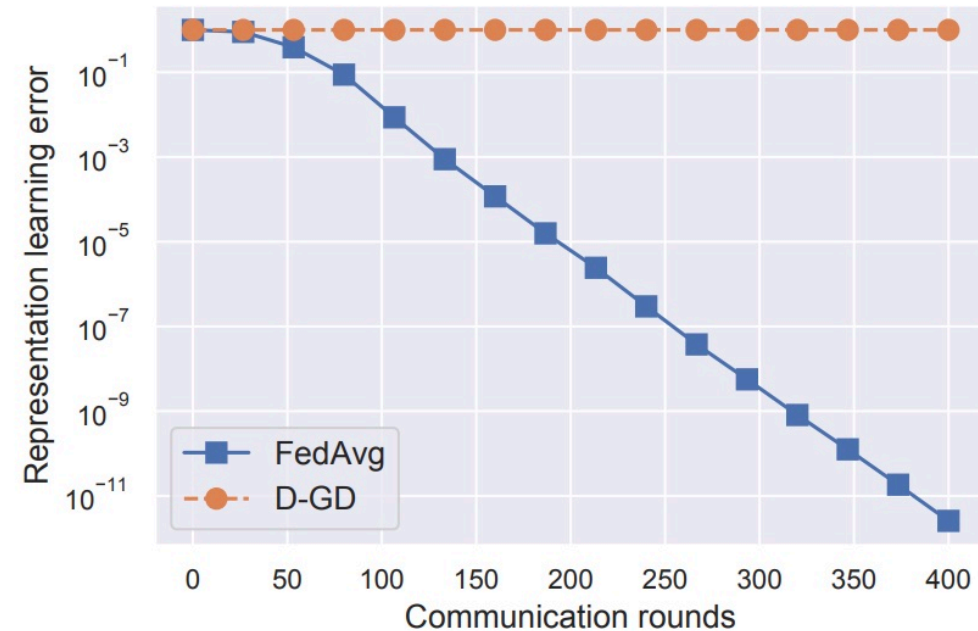
- Optimization point of view

- ▶ DGD finds a stationary point of the average loss, while FedAvg does not



- Generalization point of view

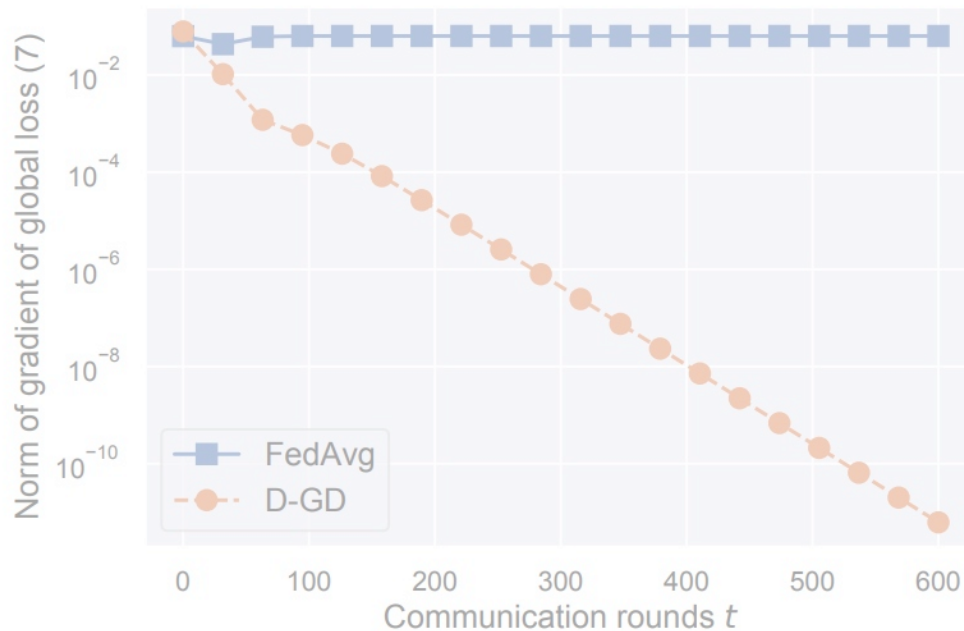
- ▶ FedAvg learns the correct representation (principal angle distance)



Important Observations and Conclusion (2/2)

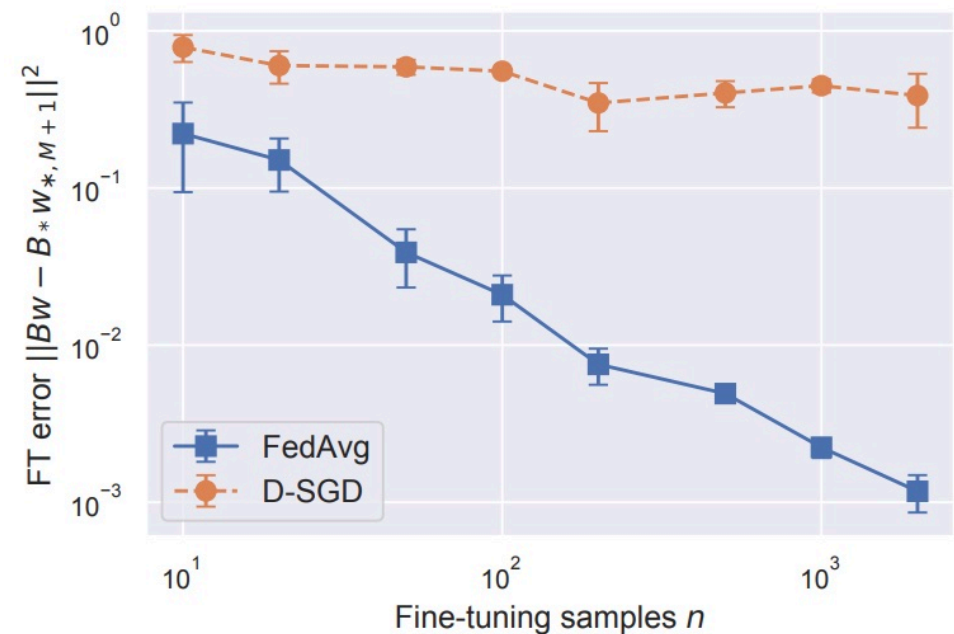
- Optimization point of view

- ▶ DGD finds a stationary point of the average loss, while FedAvg does not



- Generalization point of view

- ▶ FedAvg generalizes better (loss with new task, after fine tuning)!



Further Details

- “MAML and ANIL Provably Learn Representations”, Liam Collins, Aryan Mokhtari, Sewoong Oh and Sanjay Shakkottai. *Proceedings of the 39th International Conference on Machine Learning (ICML), Baltimore, MD, July 2022*. <https://arxiv.org/abs/2202.03483>
- “FedAvg with Fine Tuning: Local Updates Lead to Representation Learning”, Liam Collins, Hamed Hassani, Aryan Mokhtari, Sanjay Shakkottai. *To appear in the Proceedings of the Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), New Orleans, December 2022*. <https://arxiv.org/abs/2205.13692>